RAKSHAM · ROUND 1 · TRACK 03

TRUVY

Know what's real.

AI-powered trust intelligence for financial communications.

TEAM RED HAT

Waqar Akhtar · Ojus Mathur · Ujjwal Sharma

Challenge · Deepfake Detection for Financial Communications



A Familiar Voice Is No Longer Proof.

ILLUSTRATIVE SCENARIO · NOT A REAL CASE

A finance officer receives a voice note that sounds exactly like the company's CEO.

The message requests an urgent transfer to a new beneficiary. A convincing video appears to confirm the instruction. The sender discourages a callback, claiming to be unavailable.

But the real executive never authorised the transaction.

01 · FAMILIARITY

Familiarity creates trust.

Employees naturally recognise voices, faces and communication patterns. Attackers exploit this familiarity.

02 · URGENCY

Urgency suppresses verification.

Time pressure, authority claims, secrecy and callback avoidance encourage employees to bypass normal controls.

03 · DETECTION

Detection does not equal prevention.

A media detector can surface signs of manipulation, but its score alone cannot establish whether the financial instruction is legitimate.

HARM

- Fraudulent transfers
- Unauthorised beneficiary changes
- Difficult recovery
- Operational & reputational damage

The vulnerability is not simply a fake voice. It is the gap between recognising someone and verifying that they authorised an instruction.

From Detecting Deception to Verifying Intent.

01

DETECT

Identify manipulation

Analyse audio and video for signs of synthetic media or manipulation.

02

ASSESS

Weigh the context

Combine forensic findings with language confidence, pressure cues and payment context.

03

VERIFY

Route to trust

Recommend trusted-channel callbacks, human review and evidence documentation.

DIFFERENTIATION

Conventional media analysis focuses on whether content **appears manipulated**. TRUVY connects media analysis to an **operational decision** and a **verification process** · it does not rely on a single detector score.

TRUVY helps answer not only whether media looks suspicious, but what should happen before the associated payment proceeds.

Every Suspicious Instruction Has a Verification Path.

Incoming instruction
Audio or video · payment or beneficiary change
● INTAKE

↓

MEDIA ANALYSIS
- Audio forensics
- Video analysis
- Language identification
- Provenance checks where available

CONTEXT ANALYSIS
- Pressure cues
- Transaction amount
- Beneficiary changes
- Timing & communication context

↓

Risk assessment
Combine signals under explicit rules · a risk band

↓

● LOW
Continue
Proceed through normal maker-checker controls and log the case.

● ELEVATED
Callback
Require an independent callback before payment release.

● HIGH
Hold & review
Hold the payment for human review and require dual approval.

EVIDENCE Every outcome writes an **evidence record and report package**.

Critical rule: callback numbers come from the institution's trusted directory, never from the suspicious message. TRUVY does not automatically reject payments.

Multiple Signals. One Explainable Risk Assessment.

LAYER 01

Media forensics

Audio deepfake detection, sampled-frame video analysis, language identification and provenance checks.

- Audio · Video · Language · Provenance

LAYER 02

Communication analysis

Speech-to-text and structured identification of urgency, secrecy, authority claims and callback avoidance.

- Transcript · Pressure cues

LAYER 03

Financial context

Transaction amount, beneficiary history, unusual timing and communication-channel anomalies.

- Amount · Beneficiary · Timing

LAYER 04

Decision engine

Combine signals, apply explicit rules, generate an explanation and route the request for the appropriate action.

- Rules · Explanation · Routing

EXAMPLE RULE

A new or changed beneficiary above a configured threshold, combined with a pressure cue, triggers a callback **regardless of the audio detector's result**.

LOW	ELEVATED	HIGH
Normal controls.	Trusted callback.	Hold + human review.

The detector informs the decision. It does not control the decision alone.

Measure the Risk. Validate the Decisions.

VALIDATION PLAN · RESULTS PENDING

No measured results are claimed at this stage.

EQUAL ERROR RATE

•

PENDING

FALSE-POSITIVE RATE

•

PENDING

PER-LANGUAGE

•

PENDING

END-TO-END LATENCY

•

PENDING

CORRECT ESCALATIONS

•

PENDING

LANGUAGE	EER	FPR	STATUS
English	·	·	Pending
Hindi	·	·	Pending
Tamil	·	·	Pending

RESOURCES UNDER EVALUATION

SEA-Spoof | HAV-DF | IndieFake | IndicSynth | SafeEar

Language performance

Evaluate English, Hindi and Tamil samples.

Recording conditions

Compare clean audio with compressed recordings representative of messaging apps.

Decision quality

Genuine instructions, impersonation scenarios, and cases where context matters despite an inconclusive detector.

Operational performance

Measure processing latency, false-positive rates and time to resolution.

Trust Requires More Than Detection.

PRIVACY

- Prototype uses synthetic, public or consented data
- No customer credentials
- No voiceprint database
- Ephemeral media processing where feasible
- Defined retention for escalated evidence

SECURITY

- Role-based access
- Encryption in transit and at rest
- Tamper-evident audit logs
- Rate limiting
- Deployment within the institution's security boundary

HUMAN CONTROL

- No automatic payment rejection
- No fraud accusation from one weak signal
- Human review for high-risk cases
- Documented reviewer overrides
- Dual approval where required

False positive

Verification or review introduces friction, but TRUVY does not independently reject the payment.

False negative

Independent payment-context rules may still trigger verification even when the media detector fails.

TRUVY is designed to support safer decisions even when its media detector is imperfect.

Built to Fit Existing Financial Workflows.

MAKER

Prepares instruction

TRUVY

Verification gate

RECOMMENDED ACTION

- Normal controls
- Trusted callback
- Human review

CHECKER / APPROVER

Authorises release

ADOPTION PATHWAY

STAGE 01

Shadow mode

Measure recommendations without imposing additional transaction friction.

STAGE 02

Advisory mode

Present explanations and recommended actions to authorised approvers.

STAGE 03

Controlled enforcement

Introduce required verification for designated risk categories with human oversight.

SCALABILITY

- Separate media-analysis services from decision logic
- Support asynchronous processing
- Allow detectors to be replaced
- Add communication modalities through modular components

Conceptual architecture · not a claim of an existing production integration.

From Signal to Decision in 48 Hours.

48

HOURS · SIX CHECKPOINTS · ONE END-TO-END DEMONSTRATOR

00 · 06	06 · 16	16 · 28	28 · 38	38 · 45	45 · 48
Environment, data preparation, baseline detector.	Audio pipeline, language ID, transcription, video analysis.	Payment-context rules, decision engine, explanations.	Evaluation and language-aware calibration.	Callback workflow, reviewer queue, evidence package.	Fallbacks, limitations and rehearsal.
One clip produces a score.	Multiple signals for a staged scenario.	Three scenarios produce bands & reasons.	Initial validation tables.	End-to-end demonstration.	Clean final run-through.

DEMO SCENARIOS

01
Legitimate routine instruction.

02
Suspected deepfake requesting a transfer to a new beneficiary.

03
Low-confidence language scenario requiring contextual verification.

FALLBACK

If the media detector underperforms, retain the contextual risk and verification workflow, present the detector as advisory, and report the measured limitations.

CLOSING

A Familiar Voice Shouldn't Be Enough to Move Money.

TRUVY combines media analysis, contextual risk assessment, trusted verification and evidence-backed human review to help financial institutions make safer decisions before payment release.

DETECT

Identify potential media manipulation.

ASSESS

Evaluate communication signals and financial context.

VERIFY

Route suspicious instructions through trusted channels.

DOCUMENT

Preserve reasons and evidence for human review.

TRUVY

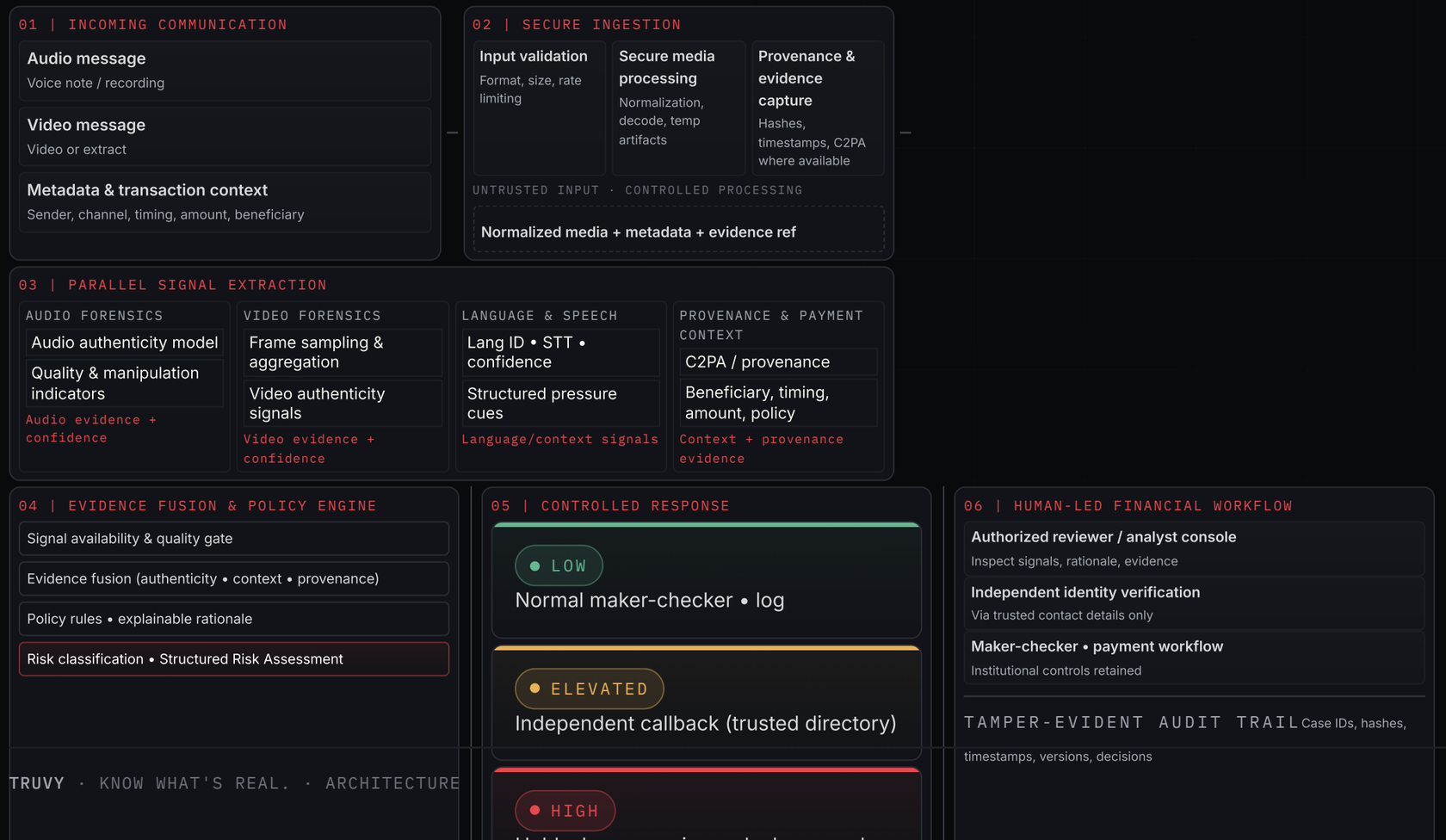
Know what's real.

TEAM RED HAT

RAKSHAM · TRACK 03

From Financial Communication to Verifiable Decision.

TRUVY combines media forensics, contextual risk analysis, and independent verification into a human-supervised workflow.



Multiple Signals. One Explainable Risk Assessment.

TRUVY separates media authenticity, contextual warning signs, and institutional policy before recommending an action.

INPUTS

Media + Context

NORMALIZATION

Secure preprocessing

Decode • quality checks

PARALLEL SIGNAL EXTRACTION

AUDIO	VIDEO	LANGUAGE & SPEECH	PROVENANCE & PAYMENT
Preprocessing • authenticity	Frame sampling • aggregation	Lang ID • STT • pressure cues	C2PA • beneficiary • timing • policy
Audio + confidence	Video + confidence	Structured cues	Context signals

RISK BAND	INTERPRETATION	RECOMMENDED ACTION	QUALIFICATION
LOW	No escalation threshold met	Normal maker-checker • log	Does not guarantee genuine
ELEVATED	Indicators justify verification	Independent callback (trusted directory)	Request for verification • not fraud
HIGH	Policy concern requiring escalation	Hold • human review • dual approval	Not automatic fraud finding

EVIDENCE PACKAGE

SIGNALS	CONTEXT	AUDIT
Scores, confidence, availability	Transcript, timing, beneficiary	Hashes, versions, rationale

INDEPENDENT VERIFICATION LOOP

- Trusted directory lookup
- Independent callback
- Authorized human decision

Never use contact from suspicious message. Final decision remains institutional.

Trust Must Be Engineered, Not Assumed.

TRUVY limits data exposure, resists common tactics, and makes model limitations visible to reviewers.

SECURITY & PRIVACY LAYERS

- Auth • RBAC • least privilege
- Encryption in transit/at rest
- Tamper-evident audit
- Minimized retention
- Versioned policies/models

No voiceprint database. Hashes, not full raw exports by default.

VALIDATION (PLANNED)

- English/Hindi/Tamil
- Clean vs compressed
- EER/FPR per condition
- Latency/failure rates

PLANNED VALIDATION · RESULTS PENDING

THREAT	POTENTIAL FAILURE	MITIGATION	RESIDUAL
Synthetic media	Evasion / subtle edits	Multi-signal + independent verification + human review	Probabilistic; cannot guarantee detection
Low-quality/compressed	Unreliable scores	Quality gate marks signals unavailable/low conf	May remain inconclusive
Callback manipulation	Fake contact	Use trusted directory only	Depends on directory integrity
False pos/neg	Business friction or missed fraud	Verification, hold, review, monitoring	Errors cannot be eliminated

— THANK YOU

TRUVY

Know what's real.

TEAM RED HAT

Waqar Akhtar · Ojus Mathur · Ujjwal Sharma